

Handout 5.3: Fiber Test Station Record

Name: _____ Date: _____ Period: _____ Station: _____

Directions: Fill in Part 1 during the slides. Take this sheet to the station for Part 2. Fill in Part 3 during the teacher demonstration. Finish Part 4 after the envelope is opened.

Station rules: one drop of water, not a puddle. Swatches stay at the station. Wet swatches go on the paper towel with the letter showing. Nothing goes in a mouth.

Word bank in other languages

English	Turkish	Portuguese	Spanish
fiber	lif	fibra	fibra
natural	doğal	natural	natural
manufactured	yapay (sentetik)	manufaturada (sintética)	manufacturada (sintética)
cotton	pamuk	algodão	algodón
wool	yün	lã	lana
silk	ipek	seda	seda
linen	keten	linho	lino
polyester	polyester	poliéster	poliéster
nylon	naylon	náilon	naílon
rayon	rayon (viskon)	raiom (viscose)	rayón (viscosa)
spandex	likra (elastan)	elastano (lycra)	elastano (licra)
absorb	emmek	absorver	absorber
stretch	uzamak (esnemek)	esticar	estirar
wrinkle	kırışmak	amassar (vincar)	arrugarse
warm	sıcak tutan	quente	cálido
melt	erimek	derreter	derretirse

Part 1: Two families of fiber

1a. Sort the eight fibers. cotton, wool, silk, linen, polyester, nylon, rayon, spandex

Natural (from a plant or an animal)	Manufactured (made in a factory)

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1b. Name the source of each natural fiber.

Fiber	Comes from
Cotton	
Wool	
Silk	
Linen	

1c. The class table. Fill in as the slides go. Use words like high, low, some, none, cheap, costly.

Fiber	Warmth	Absorbs or wicks water	Wrinkles	Stretch	Cost and care
Cotton					
Wool					
Silk					
Linen					
Polyester					
Nylon					
Rayon					
Spandex					

Part 2: The station record

Three tests on six swatches. Circle or write in every box. Ninety seconds per swatch.

Test 1, feel: rub it between finger and thumb. Circle one in each pair. **Test 2, stretch:** pull gently across, then let go. Did it come back? **Test 3, water drop:** one drop, count to ten. Soaked in, beaded up, or spread out?

Swatch	Feel: smooth or rough	Feel: warm or cool	Feel: thick or thin	Stretch: came back / stayed stretche d / no	Water: soaked in / beaded / spread	My guess at the fiber	Good for making

				stretch			
A	smooth / rough	warm / cool	thick / thin				
B	smooth / rough	warm / cool	thick / thin				
C	smooth / rough	warm / cool	thick / thin				
D	smooth / rough	warm / cool	thick / thin				
E	smooth / rough	warm / cool	thick / thin				
F	smooth / rough	warm / cool	thick / thin				

Part 3: The burn test (teacher demonstration only)

You are seated behind the line. Nobody stands up. Nobody films it.

Fiber	What the flame did (burned fast / burned slow / melted)	The smell, in one word	What was left (soft ash / crushable bead / hard bead)
Cotton			
Wool			
Polyester			

Finish the sentence: A fiber that melts is dangerous near heat because



If there is no flame today, do the wrinkle test instead. Crush each swatch in your fist for ten seconds, then open your hand. Rate the wrinkle 1 (none), 2 (some), 3 (a lot).

Swatch	A	B	C	D	E	F
Wrinkle 1 to 3						

Part 4: The reveal and the four use questions

4a. Write the real fiber next to each letter, then check your guess.

Swatch	Real fiber	Was my guess right?
A		yes / no
B		yes / no
C		yes / no
D		yes / no
E		yes / no
F		yes / no

4b. For each use, name the swatch letter you would pick and give two pieces of evidence from your own test data.

The use	My swatch	Evidence 1 (from my tests)	Evidence 2 (from my tests)
A winter hat			
A soccer jersey			
A dress shirt for a job interview			
A swimsuit			

Exit card (tear off or write on an index card)

3 things you learned about a fabric with your hands today:

1. _____ 2. _____ 3. _____

2 fibers you can now tell apart, and how:



1 question about fibers or care: